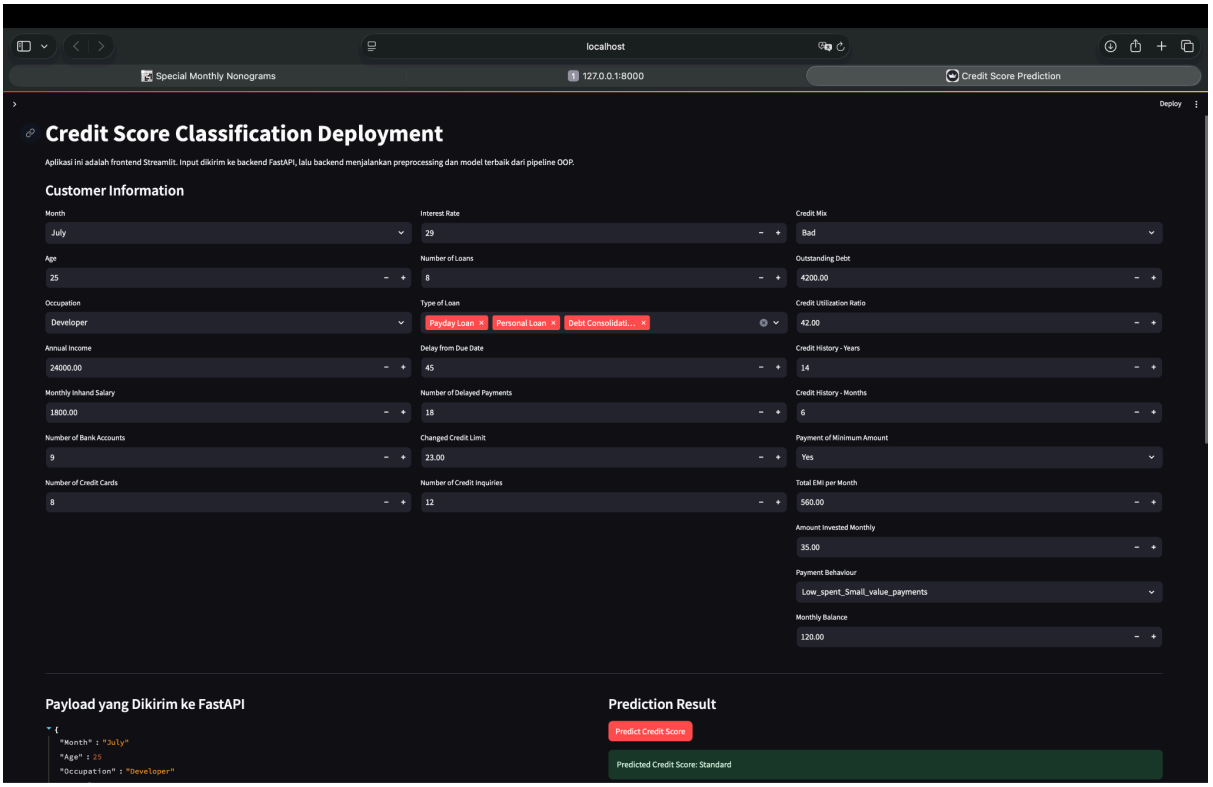
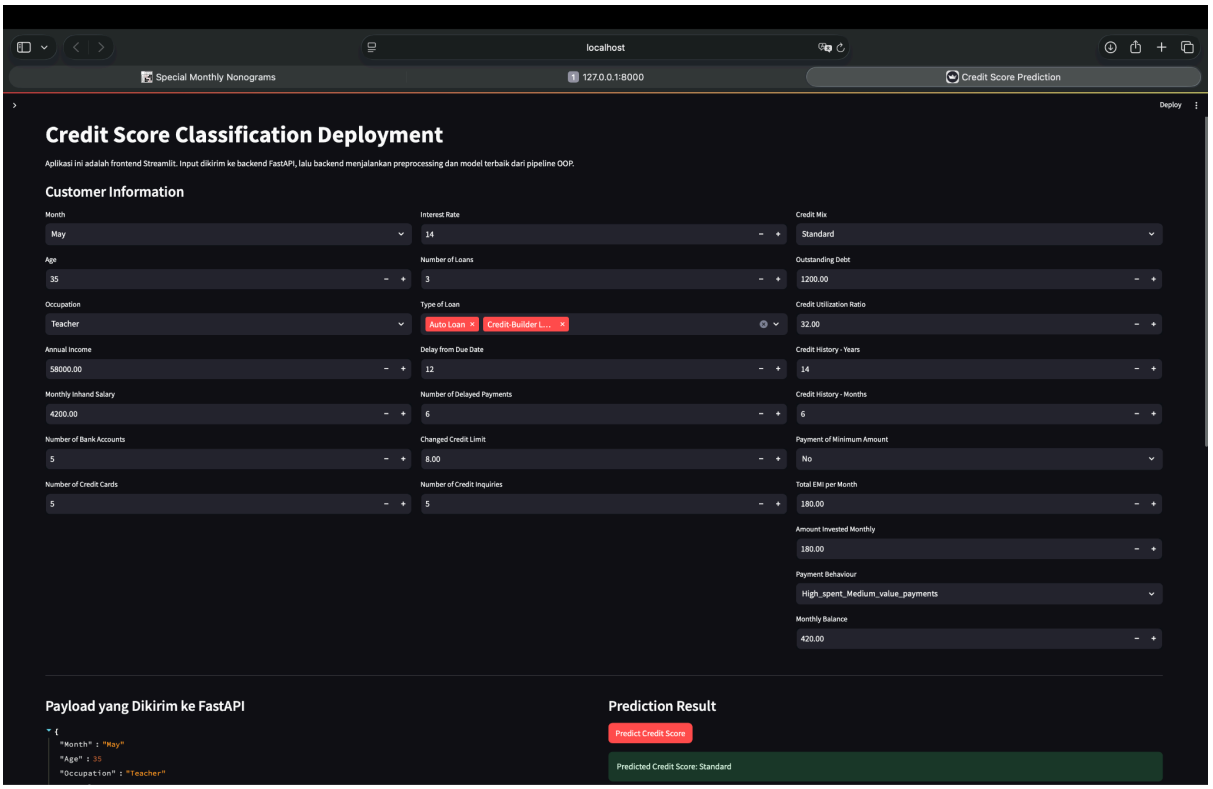
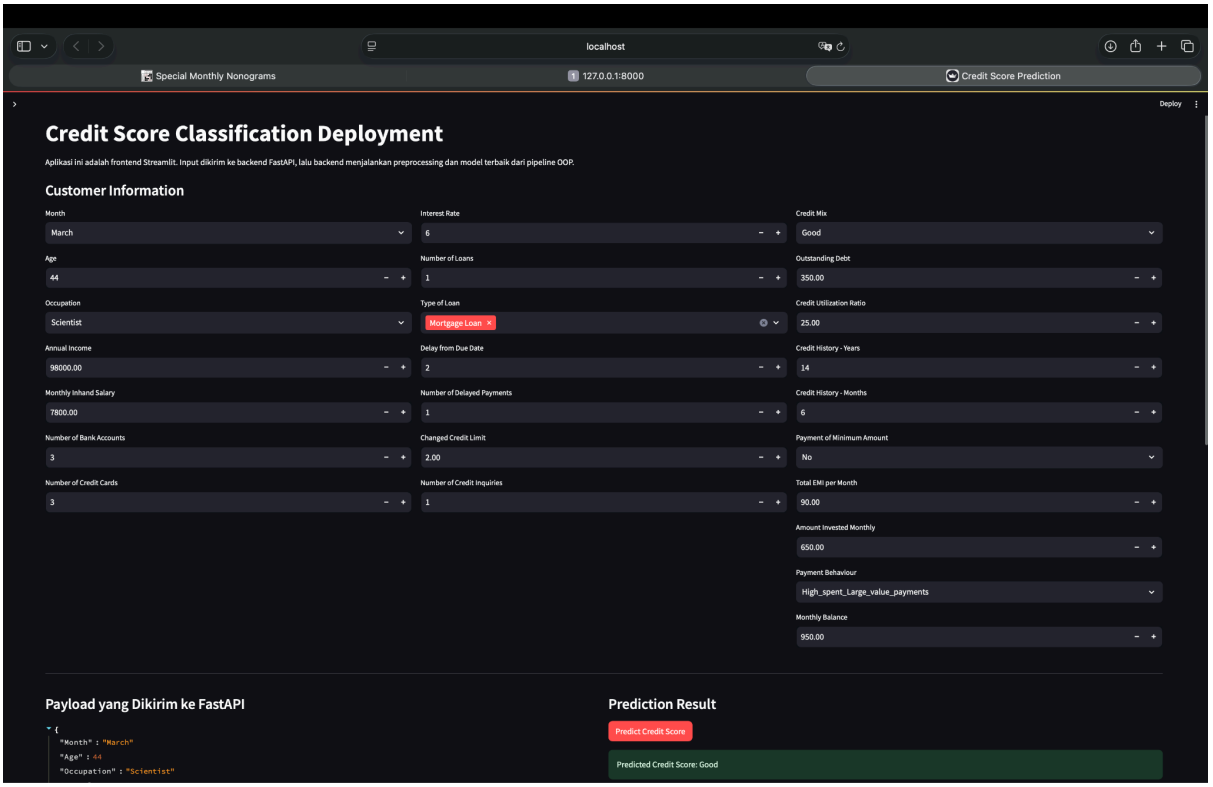
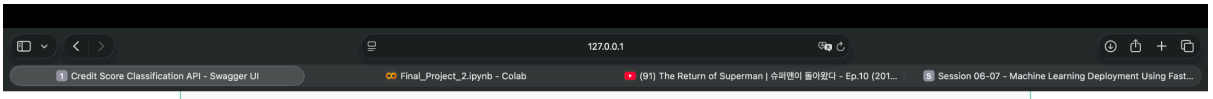
Local Deployment







FastAPI (local deployment)



Curl

```
{
  "Month": "July",
  "Age": 25,
  "Occupation": "Developer",
  "Annual_Income": 24000,
  "Monthly_Inhand_Salary": 1800,
  "Num_Bank_Accounts": 9,
  "Num_Credit_Card": 8,
  "Interest_Rate": 29,
  "Num_of_Loan": 8,
  "Type_of_Loan": "Payday Loan, Personal Loan, Debt Consolidation Loan",
  "Delay_from_due_date": 45,
  "Num_of_Delayed_Payment": 18,
  "Changed_Credit_Limit": 23,
  "Num_Credit_Inquiries": 12,
  "Credit_Mix": "Bad",
  "Outstanding_Debt": 4200,
  "Credit_Utilization_Ratio": 42,
  "Credit_History_Age": "14 Years and 6 Months",
  "Payment_of_Min_Amount": "Yes",
  "Total_EMI_per_month": 560,
  "Amount_invested_monthly": 35,
  "Payment_Behaviour": "Low_spent_Small_value_payments",
  "Monthly_Balance": 120
}'
```

Request URL

```
http://127.0.0.1:8000/predict
```

Server response

Code | Details

200

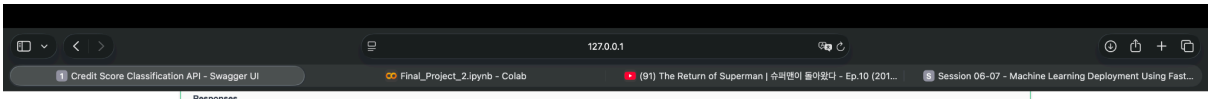
Response body


```
    "Delay_from_due_date": 45,
    "Num_of_Delayed_Payment": 18,
    "Changed_Credit_Limit": 23,
    "Num_Credit_Inquiries": 12,
    "Credit_Mix": "Bad",
    "Outstanding_Debt": 4200,
    "Credit_Utilization_Ratio": 42,
    "Credit_History_Age": "14 Years and 6 Months",
    "Payment_of_Min_Amount": "Yes",
    "Total_EMI_per_month": 560,
    "Amount_invested_monthly": 35,
    "Payment_Behaviour": "Low_spent_Small_value_payments",
    "Monthly_Balance": 120
  },
  "prediction_encoded": 2,
  "prediction": "Standard",
  "model_name": "RandomForest",
  "best_f1_weighted": 0.7453157042388165,
  "probabilities": {
    "Good": 0.052,
    "Poor": 0.368,
    "Standard": 0.58
  }
}
```


Download

Response headers

```
content-length: 835
content-type: application/json
date: Thu,02 Jul 2026 14:28:40 GMT
server: uvicorn
```



Responses

Curl

```
{
  "Month": "May",
  "Age": 35,
  "Occupation": "Teacher",
  "Annual_Income": 58000,
  "Monthly_Inhand_Salary": 4200,
  "Num_Bank_Accounts": 5,
  "Num_Credit_Card": 5,
  "Interest_Rate": 14,
  "Num_of_Loan": 3,
  "Type_of_Loan": "Auto Loan, Credit-Builder Loan",
  "Delay_from_due_date": 12,
  "Num_of_Delayed_Payment": 6,
  "Changed_Credit_Limit": 8,
  "Num_Credit_Inquiries": 5,
  "Credit_Mix": "Standard",
  "Outstanding_Debt": 1200,
  "Credit_Utilization_Ratio": 32,
  "Credit_History_Age": "14 Years and 6 Months",
  "Payment_of_Min_Amount": "No",
  "Total_EMI_per_month": 180,
  "Amount_invested_monthly": 180,
  "Payment_Behaviour": "High_spent_Medium_value_payments",
  "Monthly_Balance": 420
}'
```

Request URL

```
http://127.0.0.1:8000/predict
```

Server response

Code | Details

200

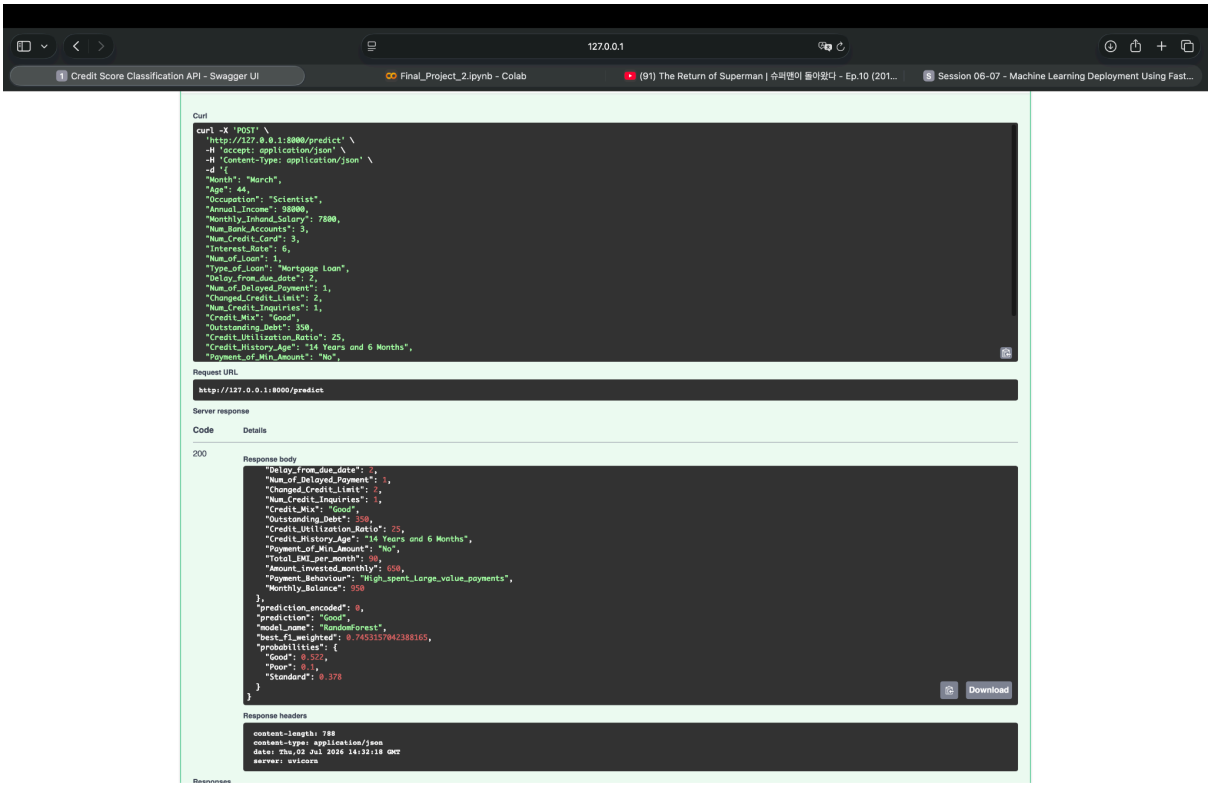
Response body


```
    "Delay_from_due_date": 12,
    "Num_of_Delayed_Payment": 6,
    "Changed_Credit_Limit": 8,
    "Num_Credit_Inquiries": 5,
    "Credit_Mix": "Standard",
    "Outstanding_Debt": 1200,
    "Credit_Utilization_Ratio": 32,
    "Credit_History_Age": "14 Years and 6 Months",
    "Payment_of_Min_Amount": "No",
    "Total_EMI_per_month": 180,
    "Amount_invested_monthly": 180,
    "Payment_Behaviour": "High_spent_Medium_value_payments",
    "Monthly_Balance": 420
  },
  "prediction_encoded": 2,
  "prediction": "Standard",
  "model_name": "RandomForest",
  "best_f1_weighted": 0.7453157042388165,
  "probabilities": {
    "Good": 0.218,
    "Poor": 0.088,
    "Standard": 0.694
  }
}
```

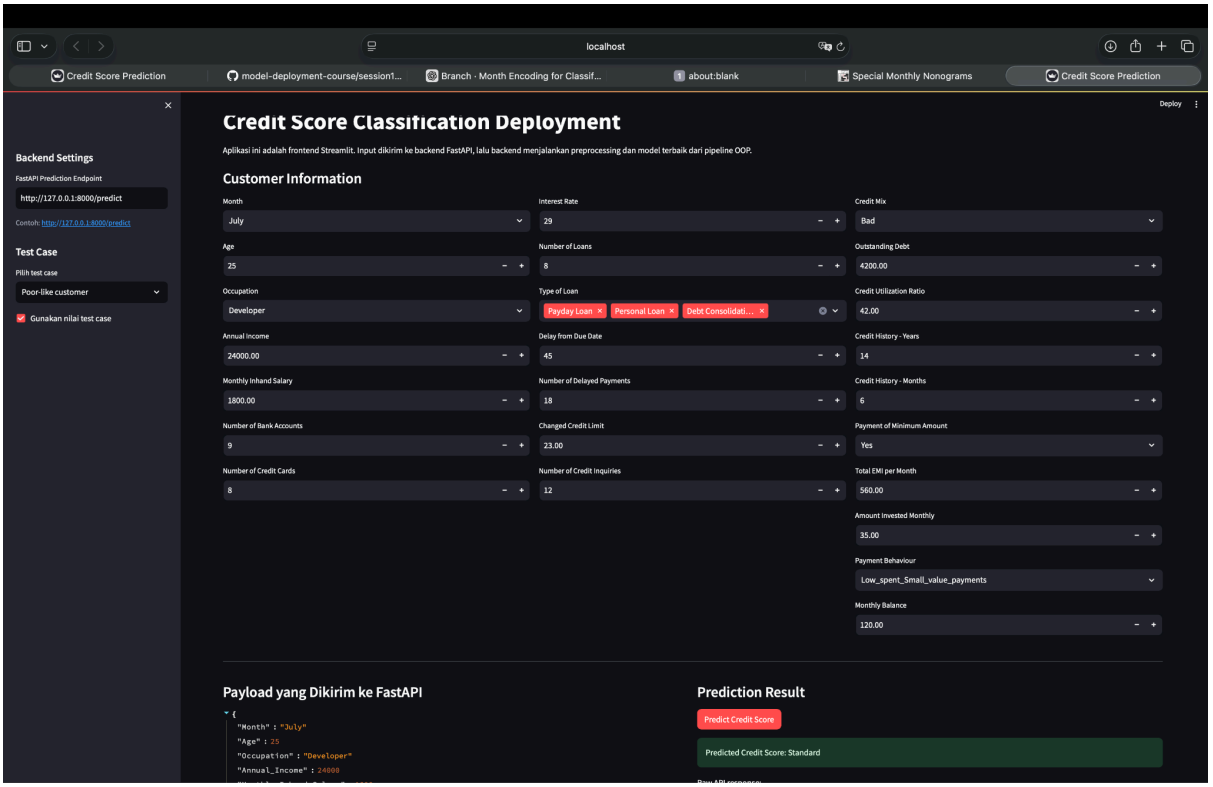

Download

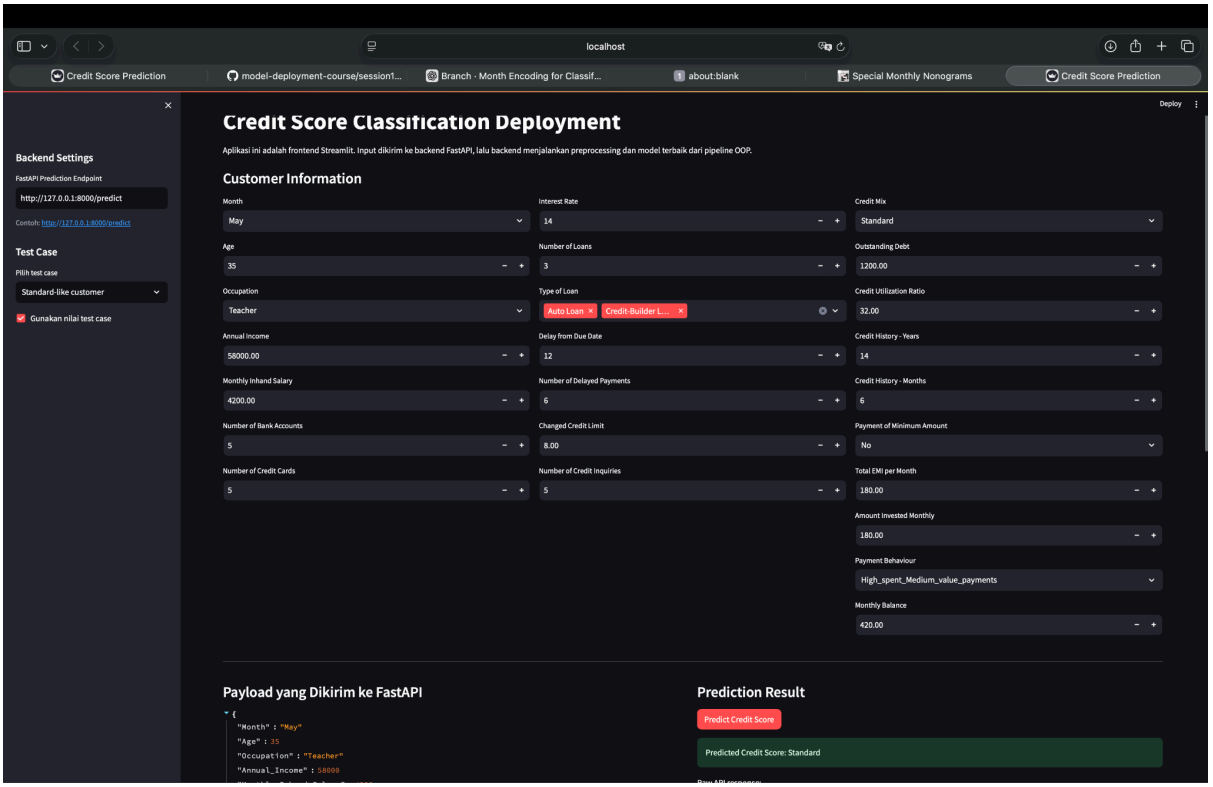
Response headers

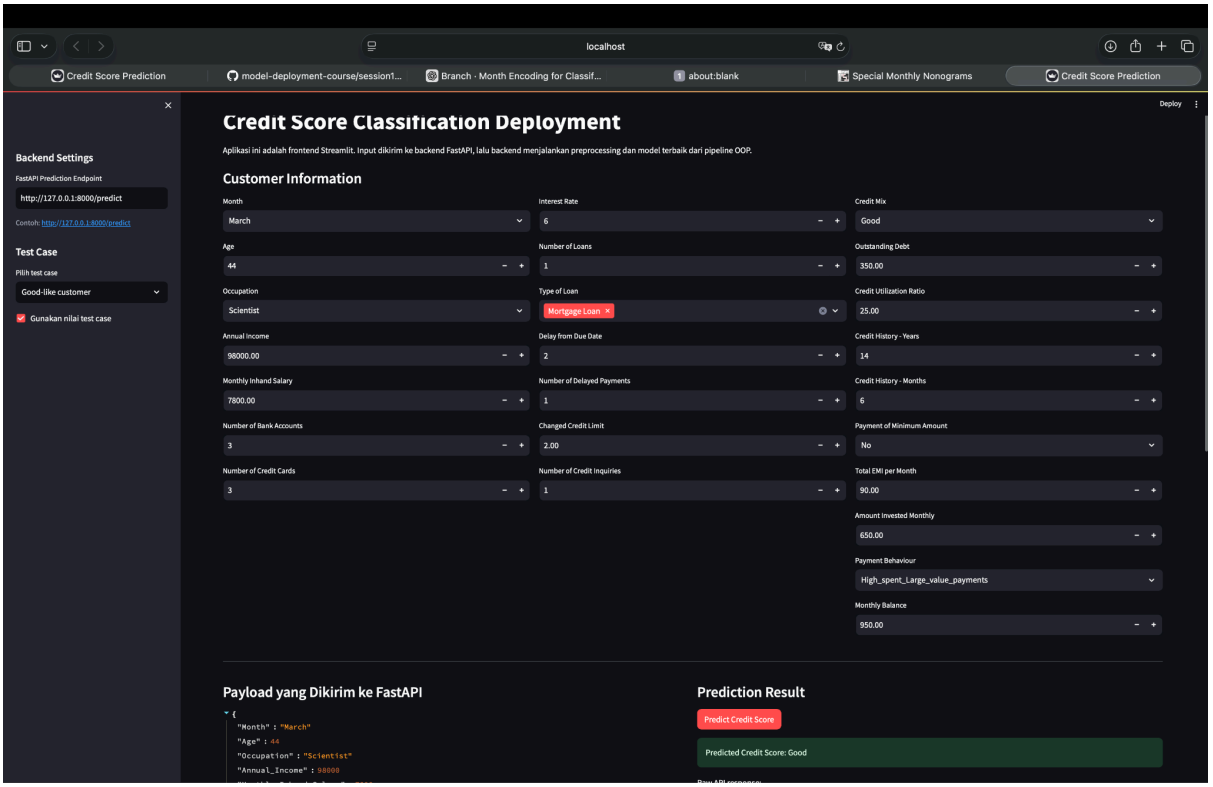
```
content-length: 816
content-type: application/json
date: Thu,02 Jul 2026 14:30:40 GMT
server: uvicorn
```



Cloud Deployment


localhost
Credit Score Prediction
model-deployment-course/session1...
Branch · Month Encoding for Classif...
about:blank
Special Monthly Nonograms
Credit Score Prediction
Deploy
Backend Settings
FastAPI Prediction Endpoint
http://127.0.0.1:8000/predict
Contoh: http://127.0.0.1:4000/predict
Test Case
Pilih test case
Poor-like customer
Gunakan nilai test case
Credit Score Classification Deployment
Aplikasi ini adalah frontend Streamlit. Input dikirim ke backend FastAPI, lalu backend menjalankan preprocessing dan model terbaik dari pipeline OOP.
Customer Information
Month
July
Interest Rate
29
Credit Mix
Bad
Age
25
Number of Loans
8
Outstanding Debt
4200.00
Occupation
Developer
Type of Loan
Payday Loan
Personal Loan
Debt Consolidati...
Credit Utilization Ratio
42.00
Annual Income
24000.00
Delay from Due Date
45
Credit History - Years
14
Monthly Inhand Salary
1800.00
Number of Delayed Payments
18
Credit History - Months
6
Number of Bank Accounts
9
Changed Credit Limit
23.00
Payment of Minimum Amount
Yes
Number of Credit Cards
8
Number of Credit Inquiries
12
Total EMI per Month
560.00
Amount Invested Monthly
35.00
Payment Behaviour
Low_spent_Small_value_payments
Monthly Balance
120.00
Payload yang Dikirim ke FastAPI
{
"Month" : "July"
"Age" : 25
"Occupation" : "Developer"
"Annual_Income" : 24000
Prediction Result
Predict Credit Score
Predicted Credit Score: Standard


localhost
Credit Score Prediction
model-deployment-course/session1...
Branch · Month Encoding for Classif...
about:blank
Special Monthly Nonograms
Credit Score Prediction
Deploy
Backend Settings
FastAPI Prediction Endpoint
http://127.0.0.1:8000/predict
Contoh: http://127.0.0.1:4000/predict
Test Case
Pilih test case
Standard-like customer
Gunakan nilai test case
Credit Score Classification Deployment
Aplikasi ini adalah frontend Streamlit. Input dikirim ke backend FastAPI, lalu backend menjalankan preprocessing dan model terbaik dari pipeline OOP.
Customer Information
Month
May
Interest Rate
14
Credit Mix
Standard
Age
35
Number of Loans
3
Outstanding Debt
1200.00
Occupation
Teacher
Type of Loan
Auto Loan
Credit-Builder L...
Credit Utilization Ratio
32.00
Annual Income
58000.00
Delay from Due Date
12
Credit History - Years
14
Monthly Inhand Salary
4200.00
Number of Delayed Payments
6
Credit History - Months
6
Number of Bank Accounts
5
Changed Credit Limit
8.00
Payment of Minimum Amount
No
Number of Credit Cards
5
Number of Credit Inquiries
5
Total EMI per Month
180.00
Amount Invested Monthly
180.00
Payment Behaviour
High_spent_Medium_value_payments
Monthly Balance
420.00
Payload yang Dikirim ke FastAPI
{
"Month" : "May"
"Age" : 35
"Occupation" : "Teacher"
"Annual_Income" : 58000
Prediction Result
Predict Credit Score
Predicted Credit Score: Standard



Aspek	Local Deployment	Cloud Deployment (AWS SageMaker)
Setup	Lebih sederhana	Lebih kompleks karena memerlukan AWS, S3, IAM, dan endpoint
Biaya	Tidak memerlukan biaya cloud	Menggunakan resource cloud sehingga perlu memperhatikan biaya
Kecepatan pengembangan	Cepat untuk debugging	Lebih lama saat deployment awal
Skalabilitas	Terbatas pada komputer lokal	Mudah diskalakan dengan instance yang lebih besar atau lebih banyak
Akses	Hanya berjalan di komputer lokal	Dapat diakses dari layanan lain selama endpoint tersedia
Maintenance	Manual	Dikelola melalui layanan AWS
Cocok untuk	Pengembangan dan pengujian	Produksi dan deployment skala lebih besar